This assignment will not be graded and is only for practice.

Key Points: 1

A map $T : V \to W$ between two vector spaces V and W is said to be a linear transformation if for any two vectors v_1 and v_2 in the vector space V and for any scalar c in $\mathbb{R}$ the following conditions hold:

$$T(v_1 + v_2) = T(v_1) + T(v_2)$$

$$T(cv_1) = cT(v_1)$$

1) Let T be a function from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined as follows: **1 point**

$$T : \mathbb{R}^2 \to \mathbb{R}^2$$
$$T(x, y) = (2x, y)$$

Choose the set of correct option(s).
[Hint: Assume $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2)$, then check the properties of linear transformation for the given function T.]

- ☐ $T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in \mathbb{R}^2$.
- ☐ $T(cv_1) = cT(v_1)$ for all $v_1 \in \mathbb{R}^2$, and $c \in \mathbb{R}$.
- ☐ T is a linear transformation.
- ☐ T is not a linear transformation.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in \mathbb{R}^2$.
$T(cv_1) = cT(v_1)$ for all $v_1 \in \mathbb{R}^2$, and $c \in \mathbb{R}$.
T is a linear transformation.

2) Let T be a function from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined as follows: **1 point**

$$T : \mathbb{R}^2 \to \mathbb{R}^2$$
$$T(x, y) = (x - 1, y + 1)$$

Choose the set of correct option(s).
[Hint: Assume $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2)$, then check the properties of linear transformation for the given function T.]

- ○ $T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in \mathbb{R}^2$.
- ○ $T(cv_1) = cT(v_1)$ for all $v_1 \in \mathbb{R}^2$, and $c \in \mathbb{R}$.
- ○ T is a linear transformation.
- ○ T is not a linear transformation.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is not a linear transformation.

Key Points: 2

A linear transformation $T : V \to W$ is said to be an isomorphism if it is bijective, i.e. the followings hold:

$T(v_1) = T(v_2)$ implies $v_1 = v_2$ (Injectivity).

For any $w \in W$ there exists $v \in V$ such that $T(v) = w$ (Surjectivity).

3) If $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$, then which of the following options is true? ***1 point***
[Hints:

Hint 1: Let $v_1 = (x_1, y_1)$, $v_2 = (x_2, y_2)$, and assume that $T(v_1) = T(v_2)$. Then check whether $v_1 = v_2$ or not.
Hint 2: Does there exist any $v \in \mathbb{R}^2$ for which $T(v) = (0, 1)$?]

- T is both one to one and onto.
- T is one to one but not onto.
- T is onto but not one to one.
- T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0

Accepted Answers:

T is neither one to one nor onto.

4) If $T : \mathbb{R}^2 \to \mathbb{R}^3$ is a linear transformation defined by $T(x, y) = (2x + 3y, 5x - y, x + 6y)$. Which of the following options is true? ***1 point***
[Hint: Does there exist any $v \in \mathbb{R}^2$ for which $T(v) = (1, 0, 0)$?]

- T is both one to one and onto.
- T is one to one but not onto.
- T is onto but not one to one.
- T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0

Accepted Answers:

T is one to one but not onto.

Key Points: 3

Standard Ordered basis of $\mathbb{R}^n$:

$$\{e_i \mid i = 1, 2, \ldots, n\},$$

where $e_1 = (1, 0, 0, \ldots, 0)$, $e_2 = (0, 1, 0, \ldots, 0)$,..., $e_n = (0, 0, \ldots, 0, 1)$. In general, $e_i = (0, 0, \ldots, 0, 1, 0, \ldots, 0)$, where 1 is at the i-th position and 0 elsewhere.

If $T(e_i)$ is known for all i, then the description of T can be known entirely. Suppose $T(e_i) = v_i$, for $i = 1, 2, \ldots, n$, then we have

$$T(x_1, x_2, \ldots, x_n) = T(x_1e_1 + x_2e_2 + \ldots + x_ne_n)$$
$$= T(x_1e_1) + T(x_2e_2) + \ldots + T(x_ne_n)$$
$$= x_1T(e_1) + x_2T(e_2) + \ldots + x_nT(e_n)$$
$$= x_1v_1 + x_2v_2 + \ldots + x_nv_n$$

5) Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation, such that $T(1,0) = (3,2)$ and $T(0,1) = (1,5)$. Which of the following options is true?
[Hint: $\{(1,0),(0,1)\}$ is the standard ordered basis of $\mathbb{R}^2$.]

1 point

- $T(x,y) = (3x + y, 2x + 5y)$
- $T(x,y) = (3x + 2y, x + 5y)$
- $T(x,y) = (5x, 6y)$
- $T(x,y) = (4x, 7y)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$T(x,y) = (3x + y, 2x + 5y)$

6) Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation, such that $T(2,1) = (3,2)$ and $T(1,3) = (0,0)$. Which of the following options is true?
[Hint: Express $(1,0)$ and $(0,1)$ in terms of $(2,1)$ and $(1,3)$.]

1 point

- $T(x,y) = \frac{1}{5}(3x, 2y)$
- $T(x,y) = (x + 1, y + 1)$
- $T(x,y) = \frac{1}{5}(9x - 3y, 6x - 2y)$
- T cannot be determined from the given information.

No, the answer is incorrect.
Score: 0

Feedback:

- $(1,0) = \frac{3}{5}(2,1) - \frac{1}{5}(1,3)$ and $(0,1) = -\frac{1}{5}(2,1) + \frac{2}{5}(1,3)$.

  Hence, $T(1,0) = \frac{3}{5}T(2,1) - \frac{1}{5}T(1,3) = \frac{3}{5}(3,2) - \frac{1}{5}(0,0) = \left(\frac{9}{5}, \frac{6}{5}\right)$

  $T(0,1) = -\frac{1}{5}T(2,1) + \frac{2}{5}T(1,3) = -\frac{1}{5}(3,2) + \frac{2}{5}(0,0) = \left(-\frac{3}{5}, -\frac{2}{5}\right)$

Accepted Answers:

$T(x,y) = \frac{1}{5}(9x - 3y, 6x - 2y)$

Key Points: 4

Let $T : V \rightarrow W$ be a linear transformation. Let $\beta = \{v_1, v_2, ..., v_n\}$ be an ordered basis of V and $\gamma = \{w_1, w_2, ..., w_m\}$ be an ordered basis of W. So each $f(v_i)$ can be uniquely written as linear combination of w_j's, where $i = 1, 2, ..., n$ and $j = 1, 2, ..., m$.

$$T(v_1) = a_{11}w_1 + a_{21}w_2 + ... + a_{m1}w_m$$
$$T(v_2) = a_{12}w_1 + a_{22}w_2 + ... + a_{m2}w_m$$
$$\vdots$$
$$T(v_n) = a_{1n}w_1 + a_{2n}w_2 + ... + a_{mn}w_m$$

The matrix corresponding to the linear transformation T with respect to the ordered bases β and γ is given by,

$$\begin{bmatrix} a_{11} & a_{12}...a_{1n} \\ a_{21} & a_{22}...a_{2n} \\ \vdots & \vdots \\ a_{m1} & a_{m2}...a_{mn} \end{bmatrix}$$

The above matrix not only depends on the linear map T, but also on the choice of the ordered bases β for V and γ for W.

It is important to note that, for a fixed choice of basis of V and W, there is a one to one correspondence between the set of all linear transformations from V to W, and the set of all $m \times n$ matrices, where $n = \dim(V)$ and $m = \dim(W)$.

7) Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$. Which of the following matrices corresponds to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain? ***1 point***
[Hint: $T(1, 0) = (1, 0)$ and $T(0, 1) = (0, 0)$]

- $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

8) Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x + y, x - y)$. Which of the following matrices corresponds to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for the domain and $\{(1, 1), (1, -1)\}$ for the co-domain? ***1 point***
[Hint: $T(1, 0) = (1, 1)$ and $T(0, 1) = (1, -1)$]

- $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0

Feedback:

• T(1, 0) = (1, 1) = 1(1, 1) + 0(1, −1), and T(0, 1) = (1, −1) = 0(1, 1) + 1(1, −1).

Accepted Answers:

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

9) Let $W = \{(x, y, z) \mid x = 2y + z\}$ be a subspace of $\mathbb{R}^3$. Let $\beta = \{(2, 1, 0), (1, 0, 1)\}$ be a basis of W. Let $T : W \to \mathbb{R}^2$ be a linear transformation such that $T(2, 1, 0) = (1, 0)$ and $T(1, 0, 1) = (0, 1)$. What will be the matrix representation of T with respect to the basis β for W and $\gamma = \{(1, 1), (1, -1)\}$ for $\mathbb{R}^2$? **1 point**

- $\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
- $\frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

10) Let $\beta = \{(1, 0), (0, 1)\}$ and $\gamma = \{(1, 1), (1, -1)\}$ be two bases of $\mathbb{R}^2$. If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is the matrix representation of a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ with respect to β for the both the domain and co-domain. What will be the matrix representation of T with respect to γ for the domain and β for the co-domain? **1 point**
[Hint: First find $T(x, y)$ and then find $T(1, 1)$ and $T(1, -1)$.]

○ $\begin{bmatrix} a+b & c+d \\ a-b & c-d \end{bmatrix}$

○ $\begin{bmatrix} a & -b \\ c & -d \end{bmatrix}$

○ $\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$

○ $\begin{bmatrix} a+b & -a-b \\ c+d & -c-d \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Feedback:

• T(x, y) = (ax+by, cx+dy), so T(1, 1) = (a+b, c+d) and T(1, −1) = (a−b, c−d).

Accepted Answers:

$\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$

Your score is: 0/10